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Developmental State or Economic Statecraft? Where, Why and How the Difference Matters

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ABSTRACT

A peculiar set of discontinuities and contradictions has recently emerged in the institutions-oriented literature that spans comparative capitalism, developmental states, and strategic techno-industrial governance. Around the globe, developmental states are reported to be both disappearing (chiefly in East Asia) and proliferating (not least in the United States). These depictions, we argue, are indicative of how the developmental state concept has become unmoored from its theoretical and historical grounding, and inadvertently politicised in scholarly debate. The concept has thus become unproductive (even if still of heuristic value). In this paper we offer a fresh way of thinking about the state's activism in both Korea and the United States. Specifically, we refurbish the idea of statecraft as it plays out in two very different national agendas, and as it is shaped by contrasting state-society relations. By paying attention to the differential international drivers and state ambitions, our analysis delivers a new and improved understanding of the character, purpose and capacities of the state in each national setting and, by implication, of their commitment and ability to confront specific challenges.

KEYWORDS

Economic statecraft;
developmental state; South
Korea; United States;
Goeconomics

A peculiar set of discontinuities and contradictions has recently emerged in the institutions-oriented literature that spans comparative capitalism, developmental states, and strategic techno-industrial governance. Around the globe, developmental states are reported to be both disappearing (chiefly in East Asia) and proliferating. On one hand you find the developmental state being 'disappeared': its contribution, its capacities, and its very existence have become widely contested – so much so that many commentators are left asking 'where is there a development state in existence today?'

On the other hand, alongside the 'disappearing' developmental state, one can also find quite the opposite tendency whereby the developmental state label is being applied more and more liberally, so that developmental states can now be found almost anywhere – both in the past and the present, from Scandinavia to the United States. In other words, it is not just the usual suspects of Japan, Korea, Taiwan, Singapore, and more lately China – or even France – that have attracted the developmental state (DS) label, but also Sweden, Finland, Denmark, Norway, Germany, Ireland, Israel, Brazil – and most recently, the United States (Block 2008, Lazonick 2008, Schrank and Whitford 2009, Wade 2017). Often included in this very lineup is one of our own works (Weiss 2014) – despite being written as a counterargument to the idea of the United States as a developmental state that pursues an 'industrial policy'.

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We see these expansive applications of the concept as problematic for two reasons. As elaborated below, they illustrate how the state's role in economic life has become politicised in scholarly debate. As a result, the developmental state concept has become contested and unproductive (even if still of heuristic value in allowing us to identify eligible cases). Flowing on from this problem, the analyst's conceptual toolbox in which to understand varieties of state activism is now found wanting. This matters when states face pressing new challenges – such as the rise of China – and the need is to understand how and why they respond (or not) in a given manner, and how well a state's response is matched to the perceived threat. By paying attention to the differential international drivers, state ambitions and state-society relations, our analysis delivers an improved understanding of the character and purpose of the state in each national setting and, by implication, of their willingness and ability to confront specific challenges.

In this paper we craft a fresh argument concerning the differential role of the state in South Korea and the United States. To this end, we repurpose the idea of statecraft as it plays out in two very different national agendas. By deploying the idea of statecraft, we open a window onto motivations and drivers: viz., why a state may be motivated to shape the technological direction within its environment. 'Statecraft' differs from industrial policy. The latter encompasses any kind of state support aimed at economy activity, regardless of objectives, drivers or motivation (e.g. job creation, company bailouts, tariff protection and tax incentives). By contrast, statecraft requires us to consider the specific international pressures and domestic enablers that may drive states to act proactively and with some consistency in the domestic high-technology arena. The payoff of a statecraft approach is the ability to distinguish the different nature, sources and consequences of techno-economic activism in each national setting. As we shall see, the patterns of activities in question fit neither the retreatist representation of South Korea, nor the developmentalist image of the United States that now vexes the DS debate. Our focus on statecraft thus provides a productive analytical tool for breaching the impasse in the DS literature. Beyond this, it sharpens our understanding of the interplay of domestic and international influences that can limit or encourage particular patterns of state activism.

Political Origins of the Developmental State Debate

The debate surrounding the developmental state has now been more than 30 years in the making but in the more recent phase of the literature, there has emerged a strong element of 'schizophrenia'. A sizeable body of work tells us that East Asia's developmental states have virtually disappeared from the scene as democratisation and globalisation advance and financial crises take their toll. At the same time we are invited to believe that the developmental state is now being propagated in many other parts of the world, from Ireland (O'Riain 2004) to Rwanda (Mann and Berry 2016) – and even to the United States (Block 2008, 2011).

The concept's recent application to the United States can be considered as an extreme case of the tendency to stretch the DS idea. After all, the comparative capitalism literature conventionally labels the United States as the quintessential Liberal Market Economy (LME) – the least likely contender for the DS mantle. It is as if we are all Developmental States now (except the East Asian pioneers!). Arguably, this proliferation is a new form of 'disappearing' the developmental state, in so far as it loses sight of what makes (or has made) this form of state so distinctive.

Clearly, we endorse neither of these two extremes of disappearance and proliferation, which are at times little more than political exercises. While its politicisation is recent (see below), the DS concept nonetheless has strong 'political' origins. The coining of the term itself has origins in a Cold War climate that blinded American analysts to the state's potentially positive economic role. In famously coining the concept, Chalmers Johnson (1982) set out to strip away the ideological blinkers that prevented US policymakers from understanding and addressing the nature of the Japanese challenge. However, while driven by the desire to influence US policymakers, Johnson's approach remained empirically and theoretically rigorous.

By fashioning the developmental state as a *contrastive* concept, Johnson sought to emphasise the fundamental goals that motivate a state's pattern of involvement in the economy. Policymakers in the goal-directed state (Japan and later, South Korea and Taiwan) targeted specific sectors for promotion because they were convinced that what a country produced would make a difference to its future prosperity. This observation led Johnson to *contrast* the developmental state with the 'regulatory state' – a state whose leaders were philosophically more inclined to let the market shape the nation's industrial structure. For Johnson, this was an orientation exemplified by his own country's approach to governing the economy. Unlike the Japanese, US policymakers did not seek to shape market outcomes. Instead, they gave priority to procedural rules over substantive results – such as ensuring that free trade and competition in the market are observed. Taken to its extreme, this 'regulatory' approach asserts that there is no economic difference between the production of silicon chips or potato chips. (an assertion famously made by the first US Bush administration when opposing what it saw as 'industrial policy' – viz. the renewal of funding for the semiconductor consortium, Sematech). The US government's apparent inactivism in the context of Japan's relentless rise sparked a heated debate over the nation's need for an industrial policy that lasted almost a decade (see Tyson 1990, Graham 1993, Faux 2009).

From Political Origins to Politicisation

Closer to our time, China's impressive technological transformation and aggressive inroads into US markets, discussed below, have helped to enliven an industrial policy advocacy that takes aim at the regulatory-cum-neoliberal state agenda. Indeed, a number of neoliberal-regulatory state critics are now advancing an empowering message: *Not only should the US have an industrial policy; it already has one – as a developmental state. We just can't see it because it is 'hidden' from view. Moreover we could make it much more coherent and effective if we cast aside the neoliberal ideology and embrace the state as an economic actor* (Block 2008, 2011, Mazzucato 2013, Wade 2017).¹ This new advocacy differs from Chalmers Johnson's. Johnson (1999, pp. 41–3) was deeply reluctant to model a developmental state (and did so only at his editor's urging), fearing that it risked a tendency to oversimplify and decontextualise experience – most fundamentally the historical drivers that motivate policy leaders and give states their *raison d'être* – and thereby overstate the potential for replication.

In light of this new advocacy, it is worth noting that Johnson's fears played out directly as anticipated in debates about the East Asian experience. In the case of South Korea, what can be called a *declinist movement* has adopted an exclusive focus on selective policy outputs and organisational changes, largely ignoring the motivations and pressures, past and present, that help keep the state economically active.² A sizeable portion of the declinist literature centred on Korea has its origins in a *Marxisant* tradition sceptical of the possibility of a state driven project, and that denies meaningful agency to the state (Pirie 2005, 2007, 2018, Chang 2009, Jessop 2016). In this school of thought, the apparent autonomy of the Korean state (and its East Asian neighbours) in its developmental heyday was little more than an historical blip made possible by a capitalist class in its infancy. In this reading, industrial maturity and capital mobility can only reveal the true nature of the state – as an instrument captive to the interests of capital. While it would be fanciful to deny the reality of this scenario in specific national settings, we wish to avoid making *a priori* claims. We also leave to one side the advocate's questions regarding what states should do. Instead we pay attention to the interplay of international and domestic influences that stimulate state activism and define its scope.

Our Approach

We contend that economic statecraft is a productive analytical tool for breaching the impasse in the DS debate. By economic statecraft we intend government initiatives designed to reach for or push the high-tech frontier in order to fend off, outflank, or move in step with clearly defined rival powers – whether such rivalry is primarily *economic* or *military*.

In particular, the concept directs our understanding to the drivers, ambitions and activities of governments in different settings. In the remainder of the paper, we elaborate the idea of statecraft as it applies to our cases and advance three propositions designed to draw out the differential nature, sources, and consequences of state activism:

- (1) The exercise of domestically focused economic statecraft can stem from radically different drivers and associated motivations (economic or military), as we demonstrate in the cases in question.
- (2) In spite of different drivers and ambitions, the exercise of statecraft (in advanced market economies) draws on similar domestic structures most productively characterised as 'governed interdependence'. The latter may or may not be harnessed for economic and/or military advantage.
- (3) Finally, distinctive state-society relations in each setting – economic nationalism in South Korea, an anti-statist ethos in the United States – enable the pursuit of state ambitions or help shape the course they take.

Economic Statecraft

Our repurposing of the concept of economic statecraft is both *necessary* and *instructive*. It is *necessary* because in conventional usage, the concept has been limited to describing a state's efforts to achieve geopolitical objectives through economic policies that are exclusively focused on the *international* arena. In exercising statecraft, states seek to influence foreign governments by deploying policy tools that provide rewards or sanctions. In this usage, the tools of statecraft (e.g. trade, aid, foreign loans, investment) are invariably outward oriented and aimed at shaping the behaviour of foreign powers. This foreign-policy focus is limited because it excludes significant ways in which geopolitical and geoeconomic forces shape economic policy initiatives focused on the *domestic* domain, not least in the high-technology arena.

For these reasons, our repurposing of the concept is also *instructive*: it draws attention to cases in which states become involved in domestic economic activities, in efforts to counter geopolitical and/or geoeconomic challenges. We do not claim to be the first study to examine domestically focused technology initiatives that respond to specific international threats. Our claim is that such initiatives should be seen as exercises in economic statecraft. Exemplary cases in our account are South Korea and the United States. As we show, each country exercises a distinctive form of economic statecraft *within the domestic arena*.

An argument which emphasises the primacy of geo-economic or geopolitical drivers and motivations is just that: an empirically substantiated set of ideas. Our argument about the primacy of one or the other driving techno activism in Korea and the US does not exclude the possibility that there *may* be cases in which geopolitics and geo-economics have more equal status. We flag such a possible case in the conclusion. But the primacy of one goal or another can only be *empirically* substantiated through primary research that speaks to the actors' intentions and, at the very least, entails understanding the context in which specific programmes are devised and the problems they are intended to solve. This brings us next to our three propositions which we develop for each national case.

From Strategic Industrial Policy to Economically Motivated Statecraft: The Case of South Korea

- (1) ***The exercise of domestically-focused, economic statecraft can stem from two different drivers (geopolitical and geoeconomic) and their associated motivations.*** In the Korean context, the drivers are principally geo-economic.

For much of the postwar period, as one of the classical 'developmental states', the Korean state practised strategic industrial policy (Amsden 1989, Woo 1991). In that period, its policy activism was

motivated by grave security concerns which in turn helped to shape and sustain the ambition to catch up economically with the advanced industrial world and attain technological independence (see Weiss and Hobson 1995, pp. 183–90). As the Cold War thawed, the security architecture of East Asia was also transformed, not least in conjunction with US–China reconciliation, and South Korea’s rapprochement with Pyongyang’s traditional allies China and the Soviet Union. Add to this picture the evolving inter-Korean dialogue from the 1980s, the two Koreas’ basic agreement on reconciliation, non-aggression and cooperation in 1991, followed by the Sunshine Policy of 1998–2003. As security scholars now attest, ‘These changes have led to a new strategic thinking in South Korea’ (Sheen 2009, p. 43). Of course, there have been periods of Pyongyang’s sabre rattling, but since the 1990s it is fair to say that for Seoul, ‘The possible collapse of North Korea is now perceived as a greater threat than a possible military attack’ (Sheen 2009, p. 43).³ So by the turn of the new century, Seoul’s overarching geopolitical imperative to secure itself against North Korea had been overtaken by the geo-economic imperative to secure its high-tech markets against China. To be clear, we are not claiming that Seoul no longer has security concerns. Of course it does. Our point is that where high-tech innovation strategy is concerned, the paramount driver is commercial.⁴

Korea’s shift in focus from ‘catching-up with advanced economies’ (developmentalism) to defending and extending its position vis-à-vis clearly defined foreign rivals (statecraft) can be dated to this time. At the turn of the century, Korea’s decades long developmental drive had borne fruit; its ‘developed country’ status was well established, and its massive conglomerate firms (the *chaebol*) had successfully leapfrogged into leadership positions in a swathe of high-tech industries, from automobiles and semiconductors to ICT.⁵ However, in the early 2000s, China’s rapid technological rise was perceived as a major threat to Korea’s future manufacturing capacity, technological autonomy and export competitiveness – long the core concerns of the country’s economic decisionmakers.

China’s advancement exposed the relative technological inferiority of Korea’s non-*chaebol* firms, long overlooked under the state’s original *chaebol*-centred developmental strategy. Consequently, the technological competitiveness of Korea’s smaller firms sorely lagged behind their German and Japanese counterparts. This left the *chaebol* dependent on German and Japanese imports for their most technologically intensive parts and components – a significant worry for Korean policymakers concerned with the technological autonomy and export competitiveness of local firms (Thurbon 2016a, pp. 100–1).

Weakness in Korea’s small-firm sector left it vulnerable to Chinese competition. The 1997 crisis made this new reality evident to Korean policymakers as the *chaebol* began relocating their manufacturing activities to take advantage of China’s newfound capabilities. From the early 2000s, the Korean government acted on the conviction that the country had only a brief window of opportunity in which to fend off the China challenge and to join the ranks of advanced countries (see Roh 2003). This goal demanded improving the technological capabilities of smaller firms.

We come to our main point: the China challenge has ushered in a new era of strategic activism in Korea, best understood as an era of economic statecraft. Its exercise is trained on rapidly upgrading the technological and export capabilities of Korea’s *non-chaebol* firms in an attempt to transform Korea from a ‘fast follower’ in existing industries to a leading power in the technology-intensive industries of the future – including intelligent robotics, medical devices, renewable energy and micro grids.⁶ These geoeconomic ambitions have motivated the techno-economy building projects of Korean administrations since 2003 (e.g. President Roh’s ‘IT-839’ initiative, Lee Myung Bak’s ‘Green Growth’ strategy, Park Geun-hye’s ‘Creative Economy’ project, and President Moon’s agenda for Korea’s ‘Manufacturing Renaissance’). To reiterate, we leave to one side an assessment of the results of these initiatives. Our purpose is to emphasise the shift in strategic focus.

In the developmental state literature, such activities are often labelled ‘industrial policy’. However, the generic term ‘industrial policy’ does little to convey the quite specific geo-economic imperatives driving these activities. Indeed, ‘industrial policy’ is now applied to such a diffuse range of activities –

from tariffs and tax concessions to corporate bailouts and job creation – that its analytical value has diminished.

Briefly contrasting the Korean and Australian approaches to supporting intelligent robotics neatly captures the rationale for the conceptual distinction we are making. Here we seek to emphasise the importance of distinguishing generic ‘industrial policy’ initiatives having no necessary external reference point or urgency or targets, from those designed specifically *to fend off, outflank, or move in step with clearly defined rival powers*. The Australian government has put in place a range of very modest supports to encourage business investment in developing and deploying this technology. But in contrast to the Koreans, Australian policymakers exhibit no serious interest in fostering the techno-industrial capabilities of local firms in these industries, or in developing their export orientation with a view to capturing foreign markets.⁷ In comparison, Korea’s ambitious national project to promote intelligent robotics is striking (Thurbon and Weiss 2019). It thus makes no sense to put Korean and Australian initiatives in the same category of ‘industrial policy’: their drivers, motivations and approaches (and thus almost certainly outcomes) are of an entirely different order. For these reasons we contend that while Australia – and perhaps many other countries – are practising ‘industrial policy’, Korea is exercising economic statecraft.

(2) In spite of different drivers and ambitions, the exercise of statecraft (in advanced market economies) draws similarly on domestic structures of public-private relations, most productively characterised as ‘governed interdependence’ (GI). In the Korean setting GI provides a critical vehicle for pursuing technological advantage.

In exercising statecraft, policymakers are today drawing only in part on state structures established during Korea’s developmental phase. The most important included an all-powerful Presidential Office with direct control over the direction of economic policymaking; a meritocratic bureaucracy responsible for designing and delivering economy-building projects and directly accountable to the President; a state-controlled banking sector, which allowed policymakers to fund economy-building initiatives against performance standards (Amsden 1989). From the 1960s to 1980s, this distinctive architecture endowed the Korean state with its renowned catch-up capabilities. A further ingredient of state capacity in this era was the more top-down partnerships forged with private sector actors, namely the chaebol. This set up has given rise to the outdated idea of the East Asian developmental state as necessarily ‘bureaucratic-authoritarian’ (see e.g. O’Riain 2004, Block 2008).

In the early 1990s, a series of new international and domestic challenges emerged that placed pressure on key aspects of this centralised ‘catch-up’ state structure. How these pressures have since played out is a complex story, elaborated elsewhere (Thurbon 2016a). The relevant point is that since the early 1980s, as the economy opened up and as climbing the technology ladder became more complex, so the ‘top down’ approach to industrial governance lost efficacy. Since then, Korean policymakers have crafted new modes of collaboration with the private sector. This collaborative arrangement is most productively understood as a relationship of governed interdependence (GI) (see Weiss 1995). It is through GI that Korean policymakers are exercising statecraft in response to China’s high-tech challenge.

To be sure, GI is not unique to the South Korean setting. It can be found wherever a government body collaborates with private actors and entities to advance its own objectives, while also setting the goals to be pursued and the rules of participation. As we shall see, GI also applies to the US case. More specifically, GI captures a relationship between the state and private actors that involves considerable negotiation, collaboration and partnering, one that is pursued for mutual benefit but is ultimately governed by public objectives. In light of this goal-orientation, GI also involves the deployment of disciplinary mechanisms to direct public-private relationships towards desired outcomes. Taking the form of both carrots and sticks, these mechanisms can include cost-sharing (giving business skin in the game) or time-limited incentives tied to performance outcome, typically exports.

The concept of GI thus differs significantly from that of ‘embedded autonomy’ (Evans 1995), which envisages a use-by date for the state’s strategic role and the government-business relations that underpin it (see Thurbon and Weiss 2019, pp. 12–3). In this ever-changing world characterised by a mature business sector and relentless competitive pressures, GI offers a more fruitful entry point to the question of how and when governments can continue to be effective in an open-economy environment – *if they so choose*.

Just as GI underpinned the state’s developmental capabilities from the 1980s, so it has become central to the effective execution of economic statecraft. However, GI evolves over time. During Korea’s developmental phase, GI centred on the government’s collaborative relationships with the chaebol. In the present era of reaching for the technology frontier, new forms of GI draw small-to-medium sized firms into the state’s strategic orbit. State actors view the creation of this broader network of firms as vital for the transition from fast followership to leadership. Nurturing small firms in advanced technology industries is a completely different game from that of promoting fast followership by large firms in existing industries. As a start, the very nature of advanced technology industries and the range of actors involved in bringing them to life demands a more inclusive and much less centralised approach. This new approach does not exclude the chaebol. Rather, it seeks to enlist them as partners in an effort to upgrade the capabilities of smaller firms and to transform Korean firms into leaders in the advanced-technology industries of the future. Sung-young Kim’s recent (2019) work on the government’s creation of ‘hybridised industrial ecosystems’ in the renewable energy arena, and the collaborative government business relations that underpin these systems, is a particularly noteworthy example. This concept builds on Kim’s earlier work illuminating the systems of GI that underpinned Korea’s transition from fast follower to innovator in the telecommunications sector (Kim 2012). In the case of intelligent robotics, we see how external pressures have renewed the interest of the chaebol in collaborating with government for mutually beneficial goals, that now include the universe of smaller firms (Thurbon and Weiss 2019). These studies shed new light on the diverse forms of governed interdependence that feed into the practice of economic statecraft.

(3) ***Distinctive state-society relations enable and shape the pursuit of those ambitions.***

Government-business relations are not the only important enabler of economic statecraft. State-society relations are no less important (even if widely overlooked in the developmental state literature).⁸ In the Korean setting, the relevant relationship is premised on the shared ideology of economic nationalism. Since the 1960s economic nationalism has bound Koreans to the state and to its transformative projects. By cultivating and manipulating economic nationalist sentiment, state actors have been able to enlist Korean citizens as vital partners in their economy-building initiatives. More specifically, by appealing to this sentiment, Korean leaders have been able not only to *legitimise* their strategic activism,⁹ but also to mobilise and direct the energies of Korean workers, consumers and the wider public towards the pursuit of their transformative objectives. Even today, despite decades of economic liberalisation and global integration, economic nationalism remains a powerful force within Korean society, and an important enabler of the state’s economic statecraft. Throughout the postwar period, leaders of all political persuasions have mobilised this ideology for strategic purposes.

Over the 1960s and ‘70s for example, authoritarian leader Park Chung Hee cultivated nationalist sentiment to both legitimise his developmental dictatorship and to channel the energies of an entire population towards the pursuit of economic objectives.¹⁰ Decades later, his democratic opponent Kim Dae Jung drew similarly on nationalist support when he came to power following the regional financial crisis. Indeed Kim revealed himself more than willing to intervene in the economy in the pursuit of his transformative agenda, overseeing the revival of strategic activism broadly and financial activism in particular (Thurbon 2016a, chapter 6). Like his authoritarianism predecessors, Kim also appealed to nationalist sentiment to mobilise Koreans behind sometimes painful

developmentally driven reform measures. And Koreans responded with major enthusiasm, famously donating their own personal gold – in the shape of wedding rings and other jewelery – to help the country avoid national default on its debts.

Since the early 2000s, Koreans have elected a series of presidents who have proven willing to engage in economy-building initiatives for transformative ends. Indeed, Koreans do not just comply with, but expect and often demand such activism. So when they perceive that the government is not doing enough to counter major economic challenges – they make their expectations felt at the ballot box. So it came to pass that when faced with rolling financial instabilities and a looming China challenge under Roh Moo-Hyun, Korea was gripped with a Park Chung Hee nostalgia (Moon 2009). Lee Myung Bak capitalised on this sentiment in his run for the Presidency in 2008, explicitly framing himself as a ‘Park Chung Hee for the Twenty-First Century’ – an enormously successful tactic. While Lee fell out of favour with voters for some of his ‘big ticket’ policy promises, many of his economy-building initiatives endure today (re-badged by successive Presidents). Public support however does have its limits; where strategic activism centres on smaller firms – or building productive relationships between firms small and large, it is widely supported. But where strategic activism is seen to exclusively benefit the chaebol, it faces serious public resistance (Thurbon 2016a, 149–50).

Evidently economic nationalism remains a powerful force in Korean society, which is well evidenced in recent research (see, e.g. Suh and Kwon 2002, Lee and Lee 2015). Most relevant for our argument are its consequences: strong support for the use of state power to advance the economic competitiveness of the nation and the livelihood of its people.¹¹ The situation is very different in the United States, to which we now turn.

Security Motivated Statecraft: The Case of the United States

(1) *The exercise of economic statecraft can stem from radically different drivers and motivations.*

Turning to the US experience we find a fundamentally different variety of statecraft with profound consequences for its military and economic power. In contrast to the Korean case, US statecraft has had (and continues to have) as its primary mission national security, not economic competitiveness. Most fundamentally, perceptions of an existential threat during the Cold War were pivotal in helping to institutionalise and diffuse a national security mindset within US politics and government.

To better grasp the kind of statecraft we are dealing with in the United States, we need to shift our focus to the relevant agents, those which comprise the so-called ‘national security state’ (NSS).¹² The NSS, understood here as an ‘innovation and enterprise engine’, gradually took shape as postwar hostilities between the two great powers intensified and as perceived threats loomed ever larger. Indeed, the core mission agencies of the NSS were created, expanded, and strengthened in response to a series of perceived geopolitical threats that began with the Cold War – driven first by military rivalry with the Soviet Union, later by the rise of Japan as a technology powerhouse, and more recently by the need to deal with multiple asymmetric threats (that 9/11 has come to symbolise), as well as newly emerging regional powers.

In order to deal with a perceived existential threat from the Soviet Union, planners in the Pentagon fastened on a strategy of technological dominance. Deciding early on that the United States could never hope to match the Soviets with sheer quantity of men and materiel, Pentagon officials instead insisted that America must rely on achieving technological superiority. So, the quality of technology, not the quantity of armaments, became the overarching goal, in the conviction that techno supremacy holds the key to sustaining US national security and global primacy. This came to be known as the ‘offset strategy’ (Perry 2003).

It was however in direct response to the Soviet threat, symbolised most starkly in the successful world-first launch of Sputnik in 1957, that the state centred on Washington developed a cluster of

defence and security-related agencies tasked with advancing the technological frontier (Stuart 2008). Responsibility for science and technology was concentrated in the hands of the mission agencies: Defence, Energy, Space, Science, Intelligence, and even Health were tasked with national security as their sole or significant mission. This national security cluster took shape as the nation's innovation engine, drawing widely on partnerships with research institutions (public and private) and industry.

As well as helping to diffuse and institutionalise the national security project, the great Sputnik moment served as a galvanising force for action in many arenas that produced lasting legacies (McDougall 1982). In addition to boosting the funding of education, science and technology development, Sputnik's successful launch (contrasted with America's failure) spurred in 1958 the creation of technology-intensive agencies like DARPA and NASA. Not least, it inspired the federal government to kick-start a modern Venture Capital industry in order to meet an 'innovation deficit' that threatened to impinge on its national security strategy. Thus in 1958, the hybrid Small Business Investment Corporation (SBIC) programme was launched (Weiss 2014, 55–59).

We emphasise the geopolitical and security logic because it goes to the heart of the official mindset; it explains why the US government has long been prepared to pour vast resources into high-risk projects that rely on public-private partnerships. It also throws light on why commerce and the NSS have become increasingly integrated (as we will see shortly). Indeed, geopolitical concerns have led policymakers to factor some commercial benefits into national security-related, high-tech programmes. But – as we shall see – the key issue is not *whether* a programme has commercial benefits – but *why* it may be designed to produce such benefits in the first place.

If there is a consensus within the NSS, it is that techno superiority is key to America's national security and its military primacy. Without that security logic, without that mindset, there would be little appetite for risk, little justification or need to motivate the private sector and harness its capabilities, and consequently a much-reduced capacity for radical innovation. More generally, without a persistent geopolitical stimulus, without the national security driver and the quest for technological superiority, the US innovation engine would not exist in anything like its present form; the United States would be like most other advanced countries. This is not to sanction a security-driven system of innovation; the US version is hardly without its own costs and distortions – not least the extent to which it has skewed wealth and income at the very apex of the pyramid.¹³ But that is quite another story.

Seen in this light, the United States is clearly a highly goal-oriented state (not simply focused on rule setting, as understood by the idea of the 'regulatory' or market-oriented state). Nevertheless, its primary, indeed overarching, goals clearly distinguish it from those driven by geo-economic pressures, as in the Korean case post-2000. For one thing, the underlying logic of Korean techno activism is uniquely different. As the developmental state's successor version it strives to gain and hold onto technological *autonomy* primarily to advance *economic* goals. By contrast, the NSS strives for technological *superiority*, with the primary goal of advancing *geo-strategic* objectives. In the American setting geopolitical pressures from a peer adversary have played a formative and enduring role in institutionalising the national security mindset. Not least, strategic imperatives have helped to shape a distinctive state capability, viz. an unusually high tolerance and appetite for risky long-range technology ventures, often with uncertain outcomes (as discussed below). It is this risk appetite that generates radical, revolutionary payoffs, or what Ruttan (2006) has called 'general-purpose' breakthrough technologies. If industrial policy could produce such outcomes, we would expect to find radical innovations as the norm throughout the industrialised world, rather than being so notably concentrated in postwar America.

- (2) **In spite of different drivers and ambitions, the exercise of statecraft (in advanced market economies) draws on public-private relationships of 'governed interdependence' (GI). In the US setting, however, GI is harnessed for technological superiority, the foundation of its military primacy.**

As already intimated, the distinctive capacity of the NSS is not *sui generis*. It relies on domestic partnerships and collaboration with non-state actors in US research institutions and private enterprise. While a handful of large defence contractors (like Honeywell, Raytheon, and Lockheed) remain important integrators of military equipment, the innovation that the NSS requires to ensure military primacy occurs largely outside the traditional defence industry sector (Bonvillian 2018). Thus, in order to gain access to emerging technologies and innovation capabilities, NSS mission agencies must partner with private firms in the commercial sector, forging relationships of governed interdependence for that purpose. While always important, this need has become more pronounced, especially as IT has become central to defence equipment and as the private sector has outpaced traditional defence suppliers in state-of-the-art technology.

Collaboration with private firms, however, is not straightforward and cannot be taken for granted. For example, even though Silicon Valley's high-tech firms gained their start and grew strong on the back of defence and space contracts through the 1950s and 1960s, their willingness to work on NSS projects has grown more tenuous as expanding opportunities and profits afforded by new commercial markets have become more attractive. The exit of technology firms in the late 1970s and early 1980s from customised chip production – the foundational technology for US military superiority – offers an early example of the problem (Heinrich 2002). Confronted with a dwindling supply base at the height of Cold War tensions with the Soviets, DoD agencies turned increasingly to designing *dual-use* applications to address the partnership problem and motivate innovative firms to collaborate.

Beyond providing the reluctant private-sector with an attractive incentive that would counter the flight from defence-related work, several considerations guide NSS interest and involvement in commercial viability. None of these accord with industrial policy objectives. These include extracting emerging technologies for national security missions, accessing a base for testing untried innovations, lowering defence costs through industrial-scale production, and assuring future supply for security purposes. 'Commercial viability' has thus become the watchword in NSS technology programmes, but not primarily for commercial reasons. The Department of Energy's \$5 billion Accelerated Strategic Computing Initiative (ASCI) – 1995–2005 – is a classic example of economic statecraft in action (Weiss 2014, pp. 110–3). While designed to produce major commercial innovations that would support a viable industry in high-end computing, its primary purpose was to meet mission-critical needs for simulation technology following the ban on nuclear testing (Larzelere 2007). From this initiative emerged numerous breakthrough innovations that have had major commercial success, especially in the areas of simulation and visualisation software.

Through relationships of governed interdependence, the NSS thus expands its commercial reach to advance its projects, while giving the private sector a stake in the outcomes.

Government-backed VC funds offer an unusual if effective mechanism to win private sector collaboration in NSS projects. In this case, an NSS agency – such as the CIA through its fund In-Q-Tel – acts as venture capitalist. In distinction to industrial policy, however, the intent is not to make a profit or boost a company's market potential, but to access and influence emerging technology in firms with advanced capabilities (Yanuzzi 2000). With more than 150 firms in its portfolio, the CIA's fund leads numerous investment rounds in Silicon Valley and takes equity in private companies to promote emerging technologies that can serve its missions – while at the same time producing commercially viable products.¹⁴

More generally, dual-use technology is also intended to avoid 'orphan' products and ensure the mission agency with a reliable source of commercial supply. Neither industrial policy nor 'spin-off', this 'spin-around' approach is strongly encouraged in NSS programs and is now widespread.¹⁵ In this way and many more commercial viability has become a core criterion of NSS investment, drawing the NSS agencies increasingly into promoting commercial innovation that can meet national security needs. Thus, we find the mission agencies not only investing in firms' technology via dual-use programmes. They are also partnering with industry in commercial production, establishing venture capital funds, spinning out companies, and taking equity positions in private enterprise. All this is in

addition to contracting with the private sector for innovations that don't yet exist, providing a ready market for those innovations (via government procurement), and devising the radical 'problem sets' for private actors to work on. In all these ways the exercise of economic statecraft has transformed the NSS innovation engine into a more commercially involved but national security-driven enterprise (Weiss 2014).

Evidently, then, to safeguard US technological superiority, NSS activities have evolved to encompass a good deal more than simply handing out funds for R&D. So, do they warrant the recent claim that the United States is a (so-called 'networked') developmental state doing industrial policy? Superficially, at least, it is tempting to conclude with the 'industrial policy' proponents that the United States is not all that different from the East Asian developmental state in pursuit of economic competitiveness.¹⁶ But that conclusion could only stand if we were to completely divorce such activities (innovation initiatives and programmes) from the underlying geopolitical context, motivation, and strategy that informs them – and not least from the state actors and institutions (security-related mission agencies) that drive them. Peering beneath the surface, we find a very different and distinctive form of statecraft: rather than designed to boost economic competitiveness – the primary objective of an industrial policy – these measures are invariably designed to sustain US global preeminence, and thus to dominate the technology space so central to its national security strategy.

In sum, we contest the claim of a 'hidden developmental state/hidden industrial policy' for four key reasons. First, as briefly summarised above, and argued at length in Weiss (2014, see esp. 174–8), an exhaustive review of the primary evidence reveals that none of the Pentagon's key motivations are consistent with the industrial policy claim, hidden or otherwise.¹⁷

Second, NSS insiders have themselves rejected the idea that they are motivated by commercial objectives. To take one example, as one of NASA's technology transfer programme executives declared, it makes little sense to get to industry policy via a national security mission: 'If you wanted to create a miniaturized heart pump, building a space shuttle main engine wouldn't be the most efficient way to go about doing that' (cited in Weiss 2014, p. 177).

Third, where the anticipated benefits to security agencies have not been obvious, defence support for commercial activities has been lukewarm and eventually withdrawn, even where the product might have had a viable commercial future, e.g. the post-Cold War Technology Reinvestment Program (TRP). As the Undersecretary of Defense for Acquisition and Technology put it to Congress in 1995 regarding the TRP, 'I have no interest ... in these dual-use programs or investing in these programs unless the net round trip to the DoD results in lower costs for our applications. Greater performance and lower cost is the objective.' (Paul Kaminski, cited in Weiss 2014, p. 45).

Finally, the claim of a hidden industrial policy assumes that Defence has licence to engage at will in a commercial industrial policy. If this were even remotely true, how then can we account for the reams of Pentagon documents concerned for decades to address the weakening state of the nation's industrial base? That concern has loomed ever larger with the hemorrhaging of US industry offshore. And the defence department has been powerless to stop it. We return to this point below.

This is not to deny that at times the federal government has tried to boost a particular industrial sector. Nor do we suggest that geo-economic pressures do not matter. But to catalyse serious action in the US setting, geo-economic concerns generally need to be intertwined with geostrategic interests. After all, the rise of a high-tech Japan, an ally, through the late 1970s and 1980s threw down the gauntlet to America's claim to technological leadership. The point is that technological leadership for the United States can never be a purely commercial matter. Through the 1980s, for example, the loss of market share to the new techno giant, Japan, compounded geo-economic pressures with geopolitical ones. As the crown jewels of US techno supremacy and military strategy, the loss of a semiconductor industry could not be allowed to continue without having a detrimental effect on NSS technology leadership. Not incidentally, the Pentagon took on the role of industry champion by sponsoring a public-private partnership known as Sematech. By providing half of all the federal funding for

the consortium, the defence sector's exercise of statecraft was paramount, despite at times being mis-specified as an industrial policy for commerce.

In sum, the critical test for serious US support of an industrial technology is national security. While some view Sematech as a story of effective (if hidden) industrial policy, careful analysis leaves little doubt as to the primacy of the Pentagon's geostrategic priorities – not economic objectives – in this ostensibly commercial enterprise. In an authoritative account, Van Atta and Slusarczyk (2012) show that Sematech was a rare instance of a partnership that sought to *integrate* defence and commercial objectives but ultimately *failed* in its dual aim. Defense needed a vibrant semiconductor capability and agreed to cost-share 50% with the industry consortium in order to pursue its own objectives. However industry and defence technology objectives were *misaligned* and Defence never achieved its goals (though industry did), hence the Pentagon eventually withdrew from the partnership. Far from operating as a 'cover' for industry, its tensions with the path taken by industry are well documented. Moreover, where similar efforts have not readily met the national security test – famously DARPA's flat panel display initiative, the otherwise effective Advanced Technology Program and the Technology Reinvestment Program (TRP) 'dual-use' initiative – the programmes have been either overturned or short-lived.

Indeed – and somewhat ironically – it is because the US did *not* pursue an industrial policy as such, that NSS techno-security activism has taken place in the context of a massive migration of US advanced manufacturing offshore (Bonvillian 2017). Although companies were initially pulled by competitive considerations – and encouraged by the tax code – they were increasingly pushed to offshore by an overbearing financial sector that favours companies with intangibles (software) rather than hardware (Schwartz 2016, Bonvillian 2018).¹⁸ This migration of technology-intensive production has left large gaps in the industrial ecosystem (Berger et al. 2013). As technology specialists observe, the innovation disconnect and loss of manufacturing capabilities offshore is causing a headache for the NSS (Fuchs and Kirchain 2010). The problem is that the US financial sector 'is increasingly leaving hard technologies by the wayside'; this weakens the ability of the NSS mission agencies to leverage the private sector for scaling up their hard technologies (Bonvillian 2017). For if innovation encompasses the full run of activities from invention to commercialisation for the market (Breznitz and Murphree 2013), by this logic, if you do away with manufacturing, then your capacity to scale up and commercialise withers accordingly (Pisano and Shih 2009, Berger et al. 2013). Whether the headache will become a migraine is still uncertain. What is clear is the growing recognition (in NSS policy and research circles) that the loss of production capability poses a growing threat to US pre-eminence in both economic and military spheres, especially in light of China's meteoric rise and stated ambition to overtake US leadership.¹⁹

(3) *Distinctive state-society relations enable and shape the pursuit of those ambitions.*

In an environment where anti-statist political norms flourish, strong conventions exist against the use of state power to achieve economic goals. Thus, while highly supportive of most things to do with defence and national security, the American political system remains highly resistant to institutionalised funding for outright commercial ventures – as dramatically highlighted with the ups and downs and ultimate termination of the (otherwise successful) Advanced Technology Program. The flipside of American 'anti-statism' (Friedberg 2000) is the valorisation of private initiative. This dual disposition helps to shape the way the state gets involved in the private sector. Adapting Friedberg (2000), anti-statism is understood not just as a dominant constitutional tendency to divide and distribute power, but most importantly as an ideological/normative mind-set that self-interested groups can also exploit to advantage. In the political arena, for example, this disposition helps to channel government involvement (the commercial activism of the national security agencies) toward a preference for hybrid organisational forms that merge public and private resources in distinctive and often intricate ways.

Rather than relying on heavy-handed top-down controls or simply counting on the private sector to foster innovation from the bottom up, the NSS agencies have carved out a distinctive approach that relies for the most part on public-private hybrids. These relationships generally take one of two forms: either a contractual relationship between NSS agencies, industry, and research institutions, (such as Carnegie-Mellon's Advanced Robot Manufacturing consortium); or a more permanent hybrid arrangement that institutionally fuses public purpose/resources/or ownership with private management/control (such as the CIA's venture fund). Neither wholly public nor wholly private, innovation hybrids are rather a distinctive fusion. Examples abound: They include, the numerous privately managed national labs known as Federally Funded Research and Development Centers (FFRDCs); the 'public interest' corporation Stanford Research Institute, so-called 'star' of Silicon Valley for its breakthrough innovations; and the much lauded (and highly successful) venture capital fund, In-Q-Tel, owned by the CIA. As an independent, non-profit hybrid – government-funded but privately operated – In-Q-Tel looks and operates just like a private company. It makes equity investments in firms with promising technology, participates on the board of directors, tracks new developments, and with a sizeable portfolio of companies, gets to play an influential role in shaping emerging technology (Weiss 2014, pp. 66–8; 72–3; 164–67).

As a consequence, anti-statism colours the way in which the state gets involved in the private sector – even when the mission is security driven. The result is what (Weiss 2014, chapter 7) has called a hybridisation tendency – a merging or fusion of public and private resources. More than a set of relationships or simple partnerships between different entities, it refers to an organisational set up in which public ownership and/or resources are combined with private management. The broader point is that in acting through these public-private entities, and through technology partnerships, the NSS has come to revolutionise the nation's technological capabilities and stimulate commercial innovation for the purpose of national defence.

Conclusion

As our starting point, we highlighted the confusion in the DS debate and identified two problems: the politicisation of the DS concept and the associated impairment of formerly productive analytical frameworks. Politicisation, we argued, involves the tendency to privilege advocacy over analysis, specifically with regard to industrial policy. Our approach leads us to conclude that industrial policy advocacy often suffers from the rationality assumption: it is as if by simply showing what has been done, or can be done, policymakers will adjust their actions accordingly.²⁰ But policy advocacy that downplays or disregards contextual constraints and drivers (whether international or domestic) is unlikely to motivate and sustain an ambitious techno-industrial strategy.

Our approach has put context at the centre: we identified the external drivers, the leadership motivations, and the domestic enablers that give states a mandate to pursue techno-industrial transformation. This approach has allowed us to unburden the DS concept and reserve it for its classically intended purpose. More to the point, it has enabled us to distinguish two different varieties of economic statecraft – one trained on achieving commercial competitiveness (South Korea), the other on securing technological superiority for national security (United States).

Despite their different objectives, each state must draw on the collaboration of the private sector. The exercise of statecraft thus involves building relationships of governed interdependence that must be constantly renegotiated in light of changing economic and/or security imperatives. Distinctive state-society relations (economic nationalism in Korea, anti-statism in the United States) also enable and shape the exercise of these different forms of statecraft.

In the Korean case, the US security umbrella has allowed the pressures of national security to take a back seat to economic priorities. Korea's goal of techno-industrial catch-up generated a strategic industrial policy which the state pursued for much of the postwar period. Having reached maturity in most of the advanced industrial sectors, the economic imperative has shifted from catching up to defending its position in the international economy, most notably against the Chinese advance.

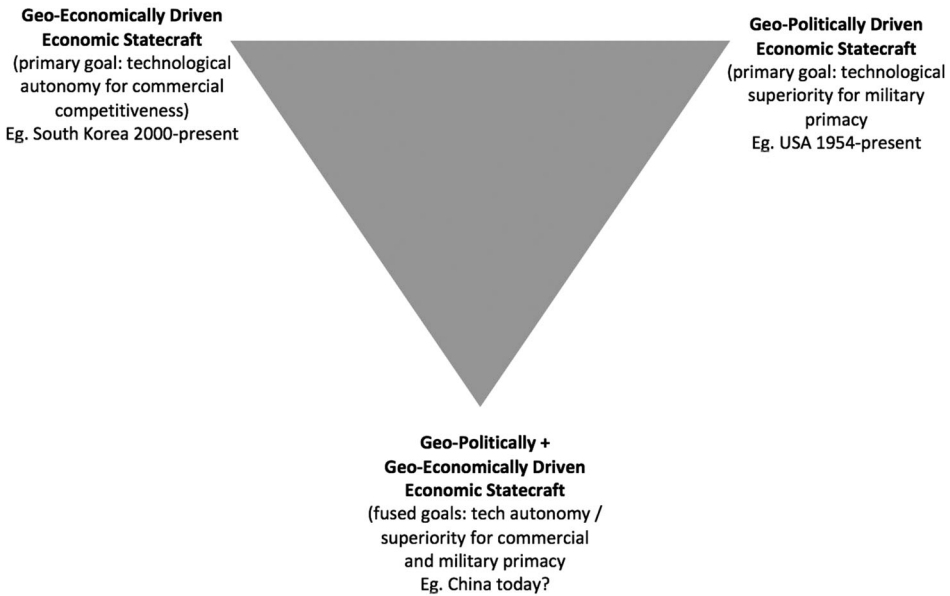


Figure 1. Three logics of domestically-oriented economic statecraft.

Source: Authors' own diagram.

In the US case, the country emerged as the world's undisputed economic and military power at the end of WWII. Economically, it remained unchallenged until the latter half of the 1970s when Japanese companies began their assault on its high-tech markets. By this stage, the Cold War had well and truly cemented techno supremacy for military primacy as America's highest priority. And given that priority, high technology could never be simply a commercial matter. So even as an ally, Japan's onslaught into US high-tech markets was perceived as a critical challenge to US national security and military primacy. By entwining geo-economic pressures with geopolitical ambitions, the rise of Japan gave impetus to commercially-friendly initiatives centred on technologies that were the life-blood of US strategic advantage (notably, semiconductors). But as Japan sank into its decade-long decline, the impetus for such initiatives lost rationale and momentum, much to the disappointment of industrial policy advocates. Yet, even when the economic pressures subsided, national security priorities remained intact, though they demanded different strategies to deal with new asymmetric threats (highlighted by the events of 9/11).

Tellingly, after nearly two decades of US obsession with the Middle East, which blunted its laser-like focus on high-tech supremacy, NSS insiders are now concerned that America's techno-military advantage has atrophied at the very time when it is facing a new peer adversary. It has taken China's recent rise as a credible economic and military threat to re-energise the US focus on high technology as a national strategic imperative (Weiss 2019).

China's rise opens up a new set of questions about domestically oriented economic statecraft, not just in the United States and Korea, but right across the globe. Among the questions for future research, we would include the following: Is the NSS model one that is applicable beyond the United States? The most likely current candidate is China. But China would seem to be a *sui generis* case given its project to achieve military-economic fusion. Indeed, in view of its stated ambition to lead in *both* military and economic arenas, this fusion offers a third distinctive statecraft logic (see Figure 1) – one that could well impart an advantage that will confound assumptions about the superiority and triumph of democratic politics. More broadly, one might ask whether China's rise is a harbinger of domestically oriented economic statecraft in other national settings. Leaving these

questions aside, we can conclude that economic statecraft is a timely new addition to the analyst's conceptual toolbox.

Notes

1. In addition to these policy advocates, the industrial policy idea has also found favour among policy critics who seek to play up the existence of a gap between the U.S. government's free-market advocacy and its actual practice (e.g. Wade 2017).
2. For an empirically grounded critique of this literature see Thurbon (2016a). For another critique of the declinist literature see Wade (2018).
3. The US think-tank RAND corporation maintains that the risk of large-scale conflict on the Korean peninsula remained relatively low from the 1960s right through to 2015, after which acceleration of the North's nuclear ambitions posed a strategic concern for South Korea (and the US), alongside the ongoing risk of state collapse (RAND 2018). Nonetheless, the South remains wholly dependent on US military leadership and operational control in response to any Northern attack, offering some insight into why commercial competitiveness concerns continue to dominate domestic debates about Korea's techno-economic policy direction.
4. Although Korea has ramped up military R&D over the past decade, analysts observe that 'South Korea's defence industrial strategy is to a great extent guided by the political [economic] ambition to increase the quantity and value of its exports of defence materiel' (Korkmaz and Rydkvist 2012, p. 92) and to establish itself as an '[arms] export powerhouse' (FPRI 2019) – rather than the (security) ambition to achieve greater technological independence in the military sphere. Although Korean leaders have occasionally expressed a desire to establish military independence, the reality is that substituting a larger proportion of US-made arms for Korean ones is likely to put interoperability with US forces at risk, compromising Korean security (ISDP 2019). This underscores our point that economic (export) concerns today trump military security ambitions in the high-tech arena.
5. On semiconductor, see Mathews and Cho (2000); on leapfrogging more broadly, see Lee (2016).
6. For case studies of techno activism at the frontier, see Kim and Thurbon (2015) and Kim (2019) on renewable energies and Thurbon and Weiss (2019) on intelligent robotics.
7. For a comparison of Australian and Korean approaches to techno-industrial governance that emphasises the differences in strategic intent, see Thurbon (2016b).
8. For an argument along these lines see Hockmuth (2018).
9. And throughout the 1960s, '70s and '80s, their authoritarianism.
10. Hockmuth (2018) provides an excellent analysis of this period, and a comprehensive review of the literature on this topic.
11. An important counter-case is Taiwan since the early 2000, where a fragmented and increasingly politicised national identity has made it far more difficult for policymakers to rally the population behind a shared national project, and thus frustrated the pursuit of strategic activism (Thurbon 2019).
12. For the detailed history and empirics see Weiss (2014).
13. See e.g. Breznitz and Cowhey (2012) on the greater social benefits of an incremental approach to innovation.
14. A typical example of this crossover between security and commerce is the CIA's 2001 lead investment in Keyhole, a company producing mapping software to enable full spectrum dominance. Keyhole's technology (used in the second Iraq war) was later acquired by Google and is now integrated as Google Earth. More recent still is InQTel's partnership with Google to back the startup Recorded Future, a company that monitors the web in real time to identify relationships and predict future actions and events (Hardy 2010).
15. The term 'spin-around' has been coined to describe the process whereby an innovation originating in or devised for a national security agency is subsequently adapted for the commercial market, and then spun back to users in the NSS (Weiss 2014, 83–4; 112–4).
16. DS 'discoverists' propose that the US is a 'networked developmental state' (NDS) because of the many agencies involved, rather than one central agency (e.g. Schrank cited in Block 2011). But the NDS could easily apply to contemporary South Korea. Their contrast of the NDS with the East Asian developmental state, only works for an earlier era when the EPB held sway – hence their nomenclature 'developmental bureaucratic state'.
17. It is interesting to note the work of Weiss (2014, Preface) who reveals that – at the beginning of her US research in 2007 – she shared the 'hidden industrial policy' view of the US, but had to discard this in light of overwhelming evidence to the contrary.
18. It is not just financialisation that has contributed to the offshoring trend. Of equal importance in the migration of production (and profits) is the role of the federal government in effectively globalising its intellectual property rules; on the blowback impact on US state capacity, see Weiss and Thurbon (2018).
19. As outlined in the document 'Made in China 2025'.
20. To those industrial policy advocates who think it simply a matter of changing hearts and minds, a policy suggestion: they might appeal to skeptics by raising the status of what they are advocating and rebranding it as 'statecraft'.

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